00400

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 2, Semester 2 (2024)

IT2070 – Data Structures and Algorithms

Duration: 2 Hours

June 2024

Instructions to Candidates:

- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 7 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.
- 10 minutes reading time is allowed.

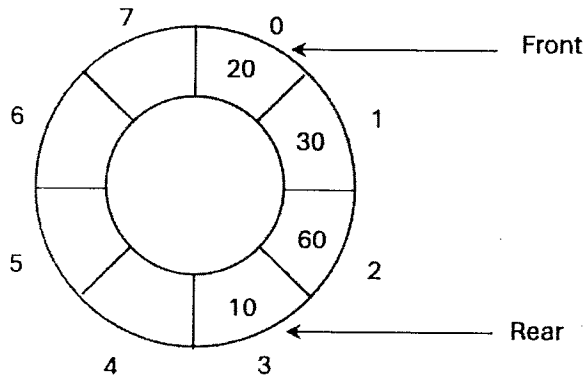
Question 1 **(25 Marks)**

a) Fill the table below considering operation principle, access point available for basic operations for Stack and Queue data structures? (4 Marks)

Data structure	Operation principle	Insertion	Deletion	Real-world Application
Stack				
Queue				

b) What is the disadvantage of a linear Queue? (1 Mark)

c) Consider the following circular queue. Draw the queue frames and indicate the values for rear, front, and count after performing below operations. (6 Marks)



i. Insert(50)
ii. Delete()
iii. PeekFront()
iv. Insert(80)
v. Insert(70)
vi. Delete()

d) "Rear - Front + 1 can be used to find the number of items in a circular queue." Do you agree with the above statement? If you agree, justify your answer using a diagram. If you do not agree, write the correct answer. (3 Marks)

e) Assume the following StackX class is implemented. You need to introduce a new method to StackX class to check whether parentheses are balanced. (8 Marks)

Write Java code segment for ***boolean isBalanced(String str)*** to take a string input from the user. Using the stack data structure your program should return true for balanced parentheses i.e. {[()]}, and false for imbalanced parentheses i.e. {{([])}].

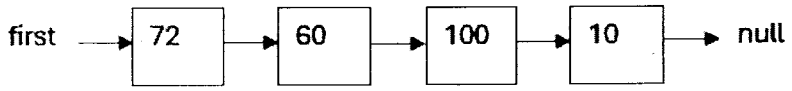
StackX
- int[] stackArray - int maxSize - int top
+void push(char c) +double pop( ) +double peek( ) +boolean isEmpty( ) +boolean isFull( )

f) Write the main program to call isBalanced(String str) and display "Parentheses are balanced" for true instances and "Parentheses are imbalanced" for false instances. (3 Marks)

Question 2 **(25 Marks)**

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a) Consider the below linked list and execute following code segments sequentially on the linked list. For each step display the output. (4 Marks)

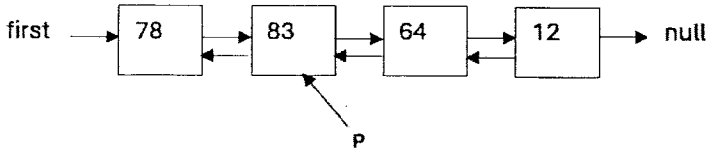


i.
```
Link temp = first;
while (temp != NULL)
{
   System.out.print(temp.ID);
   temp = temp.next;
}
```

ii.
```
first = first.next;
System.out.print(first.next.ID);
first.next = NULL;
```

b) Modify the code segment given in a) i) above to display the last link in a link list. (4 Marks)

c) Consider the following doubly link list.



i. Illustrate and write the steps to be followed if the link P is to be deleted. (3 Marks)

ii. Illustrate the steps and write the statements to be followed to insert a new link as the first link. (3 Marks)

d) What is the difference between a complete binary tree and a full binary tree? (1 Mark)

e) Consider the values given below. Arrange them into a binary search tree. (2 Marks)

250, 165, 324, 416, 278, 152, 321, 245, 130, 467, 54, 268

f) Draw the tree structure after performing following operations on the tree you have derived in 2)d). (3 Marks)

i. Delete (54)
ii. Delete (152)
iii. Delete (245)

g) Analyze and discuss the importance of a full binary tree when searching a value from a tree data structure. (5 Marks)

Question 3 **(25 Marks)**

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a) Express the T(n) of following pseudocodes using RAM model. (10 Marks)

i.
```
i = 10
while i >= 0
        if i > 5
                print i
        i = i -2
```

ii.
```
for i = 0 to n
        for j = 1 to 5
                a = i + j
                print a
```

b) "Merge Sort algorithm contains only the best case". Briefly explain this statement. (1 Mark)

c) Derive the recurrence equation for the Merge Sort best case analysis. (3 Marks)

MERGESORT (A, p, r)

1. **if** $p < r$
2. $q = \lfloor (p+r)/2 \rfloor$
3. **MERGESORT** (A, p, q)
4. **MERGESORT** (A, q+1, r)
5. **MERGE** (A, p, q, r)

d) Solve the recurrence equation obtained from 3)c) using Master Theorem and state the time complexity using Big(O) notation. (2 Marks)

$$T(n) = \begin{cases} \Theta\left(n^{\log_b a}\right) & f(n) = O\left(n^{\log_b a - \varepsilon}\right) \rightarrow f(n) < n^{\log_b a} \\ \Theta\left(n^{\log_b a} \lg n\right) & f(n) = \Theta\left(n^{\log_b a}\right) \rightarrow f(n) = n^{\log_b a} \\ \Theta\left(f(n)\right) & f(n) = \Omega\left(n^{\log_b a + \varepsilon}\right) \rightarrow f(n) > n^{\log_b a} \\ & \text{if } af(n/b) \le cf(n) \text{ for c} < 1 \text{ and large } n \end{cases}$$

e) Given Partition() is used to divide the given array in Quick Sort algorithm.

```
PARTITION(A, p, r)
1   x = A[r]
2   i = p - 1
3   for j = p to r - 1
4       if A[j] ≤ x
5           i = i + 1
6           exchange A[i] with A[j]
7   exchange A[i + 1] with A[r]
8   return i + 1
```

i. Illustrate the steps to be followed with Partition() for the following array A. (6 Marks)

A

4	6	7	9

ii. Comment on the performance of the scenario obtained in 3)e)i) and its Big(O) value. (3 Marks)

Question 4 **(25 Marks)**

a) "A heap is considered as a complete binary tree along with heap property". Briefly explain the heap property. (2 Marks)

b) In order to maintain the heap property, the following BUILD_HEAP algorithm can be used.

```
BUILD_MAX_HEAP (A)
1.    A.heap_size =A.length
2.    for i =⌊A.length/2⌋ downto 1
3.        MAX_HEAPIFY(A,i)
```

i. What is the purpose of using a for() in line 2? (2 Marks)

ii. What is the purpose of using a Heapify() in line 3? (2 Marks)

c) Use Naïve String Matching algorithm on following text and the pattern.

T = 2030403050

P= 30

i. Find the number of valid and invalid shifts for the above text and the pattern. (4 Marks)

ii. How many comparisons would occur in 3)c)? (2 Marks)

d) Use same text and the pattern in 3)c) with Rabin Carp algorithm to compute the number of valid hits and spurious hits. Consider the modulo value as 100. (4 Marks)

e) Draw the state chart transition diagram for the pattern "bba" considering the input alphabet as {a,b}. (9 Marks)

End of the paper